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Even End Sem. Examination 2022

Name of the course: B.Tech

Semester: II

Name of the Paper: Engineering Chemistry

Paper Code: TCH-201

Maximum Marks:100

Time: 3 Hour's

Note:

- All questions are compulsory.
- Answer any two sub questions among a, b and c in each main question
- Total marks in each main question are twenty.
- Each question carries 10 marks.

Q1	(10 X 2= 20 Marks)	CO 1
(a)	Draw the molecular orbital diagram of O_2 molecule and also discuss the bond order & magnetic behavior of this molecule.	
(b)	What is hydrogen bonding? Differentiate between Intermolecular and Intramolecular hydrogen bonding.	
(c)	Write a note on (i) Principle and application of Spectroscopy. (ii) Nano Scale Materials	
Q2	(10 X 2= 20 Marks)	CO 2
(a)	Discuss the zeolite method for softening of water with the help of appropriate reactions.	
(b)	A water sample on analysis give the following data: $Mg(HCO_3)_2=16.2$ mg/L; $Ca(HCO_3)_2=7.8$ mg/L; $CaSO_4=9.4$ mg/L; $MgCl_2=17.4$ mg/L ; $CaCl_2=11.1$ mg/L. Calculate the temporary and permanent hardness of water.	
(c)	Write detail note on followings: (i) Hardness in terms of $CaCO_3$ Equivalents. (ii) Reverse Osmosis method for softening of water.	
Q3	(10 X 2= 20 Marks)	CO 3
(a)	What is polymer? Discuss the preparation and applications of Nylon 6,6 and Dacron.	
(b)	What do you mean Plastics? Differentiate between Thermoplastics and Thermosetting polymers.	
(c)	Write note on followings: (i) Conducting polymer (ii) Biodegradable polymer	
Q4	(10 X 2= 20 Marks)	CO 4
(a)	What do you mean by Fuels? Classify the fuels with examples. Also write the characteristics of a good fuel.	
(b)	Write note on the followings: (i) Working of Bomb calorimeter (ii) Biogas	
(c)	Define Gross and Net Calorific Value of a fuel. The following data is obtained in the bomb calorimeter experiment: Weight of fuel=1.09 gm, water equivalent of calorimeter= 550 gm, water taken in the calorimeter= 2200 gm, observed rise in temp.=2.3°C, cooling correction= 0.0047 °C, acid correction= 60 calorie, fuse wire correction= 10 calorie. Calculate the gross and net calorific value of the fuel sample. If the fuel contains 6.5% Hydrogen. (latent heat of condensation=580 cal/gm)	
Q5	(10 X 2= 20 Marks)	
(a)	Define the term Electrode Potential and the factors affecting the electrode potential. Write the significances of Electrochemical series.	

(b)	Calculate the electrode potential of a copper wire dipped in 0.1M Cu SO ₄ at 25 degree centigrade. The standard electrode potential of $E^{\circ}_{\text{Cu}^{2+}/\text{Cu}} = +0.34 \text{ V}$.	CO 5
(c)	Write note on the followings: (i) Electrochemical theory of Corrosion (ii) Fuel cells	